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❖ Ch2_No.11

The cavity of a CO_2 laser is formed by a square plane and concave mirror. The cavity length L=50 cm, the radius of the curvature of the concave mirror R=2 m, the size of the mirror $2a$=1 cm, and the wavelength of the laser λ=10.6 μm. Calculate w_{s1}, w_{s2}, w_0, θ_0, δ_{00}^1 and δ_{00}^2.

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$L = 50\,cm$, $R_1 = +\infty$, $R_2 = 2m$, $2a = 1cm$, $\lambda = 10.6\,\mu m$

$\Rightarrow g_1 = 1 - \frac{L}{R_1} = 1 - \frac{L}{\infty} \simeq 1$

$g_2 = 1 - \frac{L}{R_2} = 1 - \frac{50\,cm}{200\,cm} = 0.75$

$\Rightarrow s1: \quad w_{s1}^2 = \frac{L \cdot \lambda}{\pi} \cdot \sqrt{\frac{g_2}{g_1 \cdot (1 - g_1 g_2)}}$

$\Rightarrow w_{s1}^2 = \frac{0.5 \times 10.6 \times 10^{-6}}{\pi} \cdot \sqrt{\frac{0.75}{0.25}}$

$w_{s1} = \sqrt{\frac{5.3}{\pi}\sqrt{3}} \times 10^{-3} \simeq 1.71 \times 10^{-3}\,m = 1.71\,mm$

$\Rightarrow s2: \quad w_{s2}^2 = \frac{L \cdot \lambda}{\pi} \cdot \sqrt{\frac{g_1}{g_2 (1 - g_1 g_2)}}$

$\rightarrow w_{s2} = \left(\frac{5.3}{\pi}\sqrt{\frac{1}{0.75 \times 0.25}}\right)^{\frac{1}{2}} \times 10^{-3} \simeq 1.97 \times 10^{-3}\,m = 1.97\,mm$

$\Rightarrow$ For this laser, since its plane and concave $\Rightarrow$ at plane

$w_0 = w_{s1} \simeq 1.71\,mm$

$\Rightarrow \quad \theta_0 = 2\frac{\lambda}{\pi w_0} = \frac{2 \times 10.6 \times 10^{-6}}{\pi \times 1.71 \times 10^{-3}} \simeq 3.94 \times 10^{-3}\,rad$

$\Rightarrow$ square plane $\rightarrow \delta_{00}^1 \; \delta_{00}^2 \rightarrow$ similar to circle

$\delta = \exp\left(-2\frac{a^2}{w^2}\right)$ we can not sure square character

$\rightarrow \delta_{00}^1 : \quad w_{s1} = 1.71\,mm , \; a = 5\,mm$

$\delta_{00}^1 \simeq \exp\left(\frac{-2 \cdot 25}{1.71^2}\right) = 3.75 \times 10^{-8}$

$\Rightarrow \delta^2_{00}$: $w_{s2} = 1.97\,mm$, $a = 5\,mm$

$$\delta^2_{00} \simeq \exp\left(\frac{-2\times 25}{1.97^2}\right) = \underline{2.54\times 10^{-6}}$$

❖ Ch2_No.13

There is a plane mirror and a concave mirror with a curvature R=1 m. How to construct a stable cavity which has the minimum far-field divergence angle of the fundamental mode? Draw the curve of relationship between the divergence angle and the cavity length L.

$R_1 = \infty$, $R_2 = 1m$

$\Rightarrow g_1 = 1 - \frac{L}{R_1} \simeq 1$, $g_2 = 1 - \frac{L}{R_2} = 1 - L$

$\Rightarrow$ stable : $0 < g_1 g_2 < 1 \Rightarrow 0 < g_2 < 1 \Rightarrow 0 < L < 1$

$\Rightarrow$ minimum $\theta_0 = \frac{\lambda}{\pi w_0} \Rightarrow w_0$: max

max $w_0 \simeq w_{s1} = \left(\frac{L\lambda}{\pi}\sqrt{\frac{g_2}{g_1(1-g_1 g_2)}}\right)^{\frac{1}{2}}$

$\rightarrow w_0 = \left(\frac{L\lambda}{\pi}\sqrt{\frac{1-L}{L}}\right)^{\frac{1}{2}} = \left(\frac{\lambda}{\pi}\right)^{\frac{1}{2}}\left(\sqrt{L(1-L)}\right)^{\frac{1}{2}}$

so we need $L(1-L)$ come to maximum

$f(L) = -L^2 + L \Rightarrow f'(L) = -2L + 1 \rightarrow$ $L < \frac{1}{2}$ $+$; $L \geq \frac{1}{2}$ $-$

$L = \frac{1}{2}$ $\quad \frac{R}{2}$

Half-confocal cavity

max of $L(1-L)$ is at $\underline{L = 0.5\,m}$

$\Rightarrow$ curve : $R = 1m$

$$\theta_0 = \sqrt{\frac{\lambda}{\pi}}\frac{1}{\left(\sqrt{L(1-L)}\right)^{\frac{1}{2}}} = \sqrt{\frac{\lambda}{\pi}}\left(\frac{1}{L(1-L)}\right)^{\frac{1}{4}}$$

$\theta_0(L)$

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$L(1-L) > 0$

$0 < L < 1$

$\theta_0(0) \rightarrow +\infty$

$\theta_0(0.5) \rightarrow \min \rightarrow \sqrt{\frac{\lambda}{\pi}}(4)^{\frac{1}{4}} \rightarrow \sqrt{\frac{2\lambda}{\pi}}$

$\theta_0(1) \rightarrow +\infty$

$\theta_{0,\min} = \sqrt{\frac{2\lambda}{\pi}}$